Rank-Preserving Calibration for Multiclass Classification

Gaurav Sood*

Abstract

Post-hoc calibration for multiclass classifiers faces two competing goals: match known class totals (label marginals) and preserve the within-class ranking of instances induced by the original model. Classical prior-correction methods and per-class logit shifts match totals but can scramble rankings via instance-specific softmax renormalization; popular multiclass calibrators (temperature, matrix/vector scaling, Dirichlet) reduce calibration error but do not enforce rank preservation. We formulate calibration as Euclidean projection onto the intersection of a row-simplex set and a columnwise isotonic-plus-sum set, where isotonic order within each class is defined by the model's pre-calibration scores. We provide a Dykstra alternating-projection solver and an ADMM splitting; the column subproblem admits an exact one-pass PAV solution followed by a scalar shift to satisfy the column sum constraint. Nearly-isotonic variants trade small, controlled rank violations for calibration gains. Experiments on neural and tabular benchmarks show that the method matches target totals while preserving within-class orderings and maintaining competitive ECE/Brier scores. Our Python implementation is open source.[1]

1 Introduction

Calibration of probabilistic classifiers is essential for decision-making where predicted probabilities should reflect empirical frequencies. In many applications, from health to finance to survey weighting, practitioners must both adjust predictions to match known class totals and maintain the model's within-class discrimination, the relative ordering of instances within each class.

*Corresponding author: contact@gsood.com

[1] `https://github.com/finite-sample/rank_preserving_calibration`

In the multiclass setting ($J \geq 3$), naive per-class logit shifts (or label-prior corrections) can change instance orderings within a class because the softmax renormalization term varies by instance. Standard multiclass calibrators such as temperature scaling [Guo et al., 2017] and Dirichlet calibration [Kull et al., 2019] improve calibration but do not guarantee order preservation. Label-shift and prior-correction methods [Saerens et al., 2002, Lipton et al., 2018] match marginals but can induce the same ranking problem.

1.1 The Rank-Scrambling Problem

Given original probabilities p_{ij} and per-class multiplicative weights w_j, the adjusted probabilities

$$q_{ij} \;=\; \frac{w_j p_{ij}}{\sum_{k=1}^{J} w_k p_{ik}}$$

depend on the instance-specific denominator. Hence $p_{i_1 j} > p_{i_2 j}$ need not imply $q_{i_1 j} > q_{i_2 j}$.

Worked example (3 classes, two instances).

$$p_{1,\cdot} = (0.51,\, 0.48,\, 0.01), \quad p_{2,\cdot} = (0.49,\, 0.02,\, 0.49), \quad w = (1,\, 100,\, 1).$$

Then $q_{11} = 0.51/(0.51 + 48 + 0.01) \approx 0.0105$ while $q_{21} = 0.49/(0.49 + 2 + 0.49) \approx 0.1644$, reversing the original class-1 order.

1.2 Our Approach

We project the original probability matrix onto the intersection of the row-simplex and per-column isotonic-plus-sum sets, where isotonic order is the total preorder induced by the original scores for that class. This formulation guarantees rank preservation and exact total matching by construction. We develop two solvers: a Dykstra alternating-projection method [Dykstra, 1983] and an ADMM splitting. The column projection in both cases is solved exactly by a single PAV pass followed by a scalar shift to match the target column sum. We also provide nearly-isotonic variants with ϵ-slack and hinge-penalty relaxations that allow practitioners to tune the calibration-discrimination tradeoff. Across neural and tabular classifiers, the method matches targets while preserving within-class weak orders and maintaining competitive calibration metrics.

2 Related Work

Label/prior shift and matching marginals. When deployment priors differ from training priors, the EM update of Saerens et al. [2002] and black-box shift estimation [Lipton et al., 2018] adjust predicted posteriors to match new class totals without refitting. These methods are effective for marginals but apply instance-specific renormalization that can reverse within-class orderings when $J \geq 3$.

Multiclass calibration. Temperature scaling and class-wise linear maps (vector/matrix scaling) [Guo et al., 2017] and Dirichlet calibration [Kull et al., 2019] are widely used but do not enforce within-class ranking constraints. Recent work explores ranking-aware criteria [Ma and Blaschko, 2021] and normalization-aware isotonic approaches for multiclass settings [Arad and Rosset, 2025]; neither simultaneously enforces rank preservation and matches given class totals.

3 Problem Formulation

Let $P \in \mathbb{R}_+^{N \times J}$ be the matrix of predicted probabilities, with rows on the probability simplex. We seek $Q \in \mathbb{R}_+^{N \times J}$ satisfying three constraints: each row sums to one ($\sum_{j=1}^{J} Q_{ij} = 1$ for all i, $Q_{ij} \geq 0$); each column sums to its target ($\sum_{i=1}^{N} Q_{ij} = T_j$); and each column is isotonic in the order defined by the original scores ($Q_{\cdot j}$ is nondecreasing when rows are sorted by $P_{\cdot j}$, with ties respected). We solve the projection

$$\min_{Q} \ \frac{1}{2}\|Q - P\|_F^2 \quad \text{s.t.} \quad Q \in S_{\text{row}} \cap I_{\text{cols}}. \tag{1}$$

The intersection is nonempty whenever $\sum_{j=1}^{J} T_j = N$ and $0 \leq T_j \leq N$ for all j: the constant-column matrix $Q_{ij} = T_j/N$ lies on the row simplex, matches the column totals, and is isotone (constant) in each column.

4 Algorithms

4.1 Dykstra's Alternating Projection Algorithm

For the row projection Π_{row}, we use a fast probability-simplex projection per row with $O(J \log J)$ time via sorting [Duchi et al., 2008] or Condat's variant [Condat, 2016].

Algorithm 1 Dykstra's method for rank-preserving calibration

Require: $P \in \mathbb{R}^{N \times J}$, targets $T \in \mathbb{R}^J$, tolerance ε, max_iters

Initialize: $Q^{(0)} = P$, $U^{(0)} = 0$, $V^{(0)} = 0$
for $k = 1$ to max_iters **do**
  $Y^{(k)} = Q^{(k-1)} + U^{(k-1)}$ (add correction)
  $Z^{(k)} = \Pi_{\text{row}}(Y^{(k)})$ (project each row onto the simplex)
  $U^{(k)} = Y^{(k)} - Z^{(k)}$ (update correction)
  $W^{(k)} = Z^{(k)} + V^{(k-1)}$ (add correction)
  $Q^{(k)} = \Pi_{\text{iso,sum}}(W^{(k)}, T)$ (per-column isotone + fixed sum)
  $V^{(k)} = W^{(k)} - Q^{(k)}$ (update correction)
  if $\|Q^{(k)} - Q^{(k-1)}\|_F / \max(1, \|Q^{(k-1)}\|_F) < \varepsilon$ **then**
    break
  end if
end for
return $Q^{(k)}$

4.2 Exact Column Projection: Isotone + Sum Constraint

For each column j, let $w \in \mathbb{R}^N$ be the current column vector arranged in the fixed order that sorts $P_{\cdot j}$ once and for all (stable, ties respected). We seek

$$\min_{x \in \mathbb{R}^N} \tfrac{1}{2}\|x - w\|_2^2 \quad \text{s.t.} \quad x_1 \le \cdots \le x_N, \ \sum_{i=1}^{N} x_i = T_j.$$

Algorithm 2 Column projection $\Pi_{\text{iso,sum}}$ (single PAV + scalar shift)

Require: Column $w \in \mathbb{R}^N$ in fixed score order, target sum T

$y \leftarrow \text{PAV}(w)$
$c \leftarrow (T - \sum_i y_i)/N$
return $y + c \cdot \mathbf{1}$

Lemma 4.1 (Isotonic projection with sum constraint)**.** *Let $C = \{x \in \mathbb{R}^N : x_1 \le \cdots \le x_N\}$ be the isotone cone and $H = \{x : \mathbf{1}^\top x = T\}$ be the sum-constraint hyperplane. If $y = \Pi_C(w)$ denotes the isotonic projection of w, then*

$$\Pi_{C \cap H}(w) = y + \frac{T - \mathbf{1}^\top y}{N}\mathbf{1}.$$

Proof. The isotone cone C is translation-invariant along $\mathbf{1}$: for any $\lambda \in \mathbb{R}$, $x \in C$ if and only if $x + \lambda\mathbf{1} \in C$, so $\Pi_C(w - \lambda\mathbf{1}) = \Pi_C(w) - \lambda\mathbf{1}$. The

Lagrangian for the sum-constrained projection

$$\min_{x \in C} \frac{1}{2}\|x - w\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^\top x = T$$

introduces a single scalar multiplier μ. The optimality conditions yield $x^\star = y + \mu\mathbf{1}$ where $y = \Pi_C(w)$. The constraint $\mathbf{1}^\top x^\star = T$ gives $\mu = (T - \mathbf{1}^\top y)/N$. □

Nonnegativity is not enforced in this subproblem; it is handled by the subsequent row-simplex projection.

Complexity. Per Dykstra iteration: row projections cost $O(NJ \log J)$; column projections are $O(NJ)$ given pre-sorted order. The orders per column are computed once upfront in $O(JN \log N)$.

4.3 ADMM Formulation

Introduce $Q = R = S$, with $R \in S_{\text{row}}$ and $S \in I_{\text{cols}}$:

$$\min_{Q,R,S} \frac{1}{2}\|Q - P\|_F^2 \quad \text{s.t. } R \in S_{\text{row}},\ S \in I_{\text{cols}},\ Q = R = S.$$

Standard ADMM updates apply [Boyd et al., 2011]. The R-update is the row-simplex projection; the S-update is Algorithm 2. Warm-starts reuse PAV block structure and previous shifts.

5 Nearly Isotonic Variants

Epsilon-slack constraints. Replace $x_{i+1} \geq x_i$ with $x_{i+1} \geq x_i - \epsilon$. Let $\tilde{x}_i = x_i + i\epsilon$. Then $\tilde{x}$ is standard isotone, and the sum constraint becomes $\sum_i \tilde{x}_i = T + \epsilon \sum_i i$. Apply Algorithm 2 to $\tilde{w}$ with target $\tilde{T}$, then recover $x_i = \tilde{x}_i - i\epsilon$.

Hinge-penalty approach. Add $\lambda \sum_i \max(0, x_i - x_{i+1})$ to the objective. The proximal step can be implemented via splitting: a standard isotonic projection substep plus a shrinkage on violations. We provide a practical solver in the package documentation.

5.1 The Fundamental Tradeoff

Without rank constraints, the natural KL-divergence projection onto the row-simplex constraint with exact column totals has the closed-form solution

$$q_{ij} = \frac{p_{ij} b_j}{\sum_k p_{ik} b_k},$$

where b_j are multiplicative weights chosen to satisfy $\sum_i q_{ij} = T_j$. This is the classical multiplicative reweighting form used in label-shift correction [Saerens et al., 2002, Lipton et al., 2018]. As the rank-scrambling example in Section 1.1 demonstrates, this exact reweighting can reverse within-class orderings because the instance-specific denominator $\sum_k p_{ik} b_k$ varies by instance, and in multiclass settings this variation can overwhelm the numerator's ordering.

One cannot in general simultaneously achieve the KL-optimal adjustment matching target priors, exact column-sum matching, and exact rank preservation. Our strict Euclidean projection (1) sacrifices KL-optimality to guarantee total matching and rank preservation. The nearly-isotonic variants allow controlled rank violations to move closer to the KL-optimal solution when small violations are acceptable. We quantify this tradeoff empirically below.

6 Theoretical Notes

Convergence. For closed convex sets, Dykstra's method converges to the Euclidean projection onto the intersection [Bauschke and Borwein, 1994]. Rates depend on geometry: polyhedral regularity can yield linear or finite convergence in special cases. We report iteration counts and primal residuals in experiments. ADMM convergence is standard for strongly convex objectives with closed convex constraints [Boyd et al., 2011, Tibshirani, 2017].

Rank preservation. Because column isotonicity is enforced in the fixed order induced by $P_{\cdot j}$ with stable ties, the calibrated $Q_{\cdot j}$ preserves the original weak order (ties may remain ties).

7 KL-Divergence Calibration

The preceding sections focused on Euclidean projection. We now develop a unified Bregman framework that encompasses both Euclidean and KL-divergence calibration, connecting to classical matrix scaling and IPF.

7.1 Bregman Umbrella Framework

We present a unified framework for rank-preserving calibration using Bregman divergences, which includes both Euclidean (squared loss) and KL (relative entropy) as special cases.

Let $D_\phi(Q, R) = \sum_{i,j} [\phi(Q_{ij}) - \phi(R_{ij}) - \nabla\phi(R_{ij})(Q_{ij} - R_{ij})]$ denote the Bregman divergence generated by a strictly convex function ϕ.

Theorem 7.1 (Bregman KKT Umbrella)**.** *Consider the penalized rank-preserving calibration problem:*

$$\min_{Q \in \mathcal{C}(T)} D_\phi(Q, R) + \lambda V_A(Q) \tag{2}$$

where $\mathcal{C}(T) = \{Q : Q\mathbf{1} = \mathbf{1},\, Q^\top \mathbf{1} = T,\, Q \geq 0\}$ *is the transportation polytope with row-simplex and column-sum constraints, and* $V_A(Q) = \sum_{j=1}^{J} \sum_{r=1}^{N-1} \omega_{jr} [(DS_j Q_{\cdot j})_r]_+$ *is the weighted rank violation penalty with* S_j *permuting column* j *by anchor* A*.*

The KKT conditions for (2) *are:*

$$\nabla\phi(Q^\star_{ij}) = \nabla\phi(R_{ij}) - \nu_i - \mu_j - [S_j^\top D^\top \alpha_j]_i \tag{3}$$

where ν_i *and* μ_j *are dual variables for row and column constraints respectively, and* $0 \leq \alpha_{jr} \leq \lambda\omega_{jr}$ *are the rank violation multipliers.*

Special Cases:

(a) ***Euclidean*** *(*$\phi(x) = \frac{1}{2}x^2$*, so* $\nabla\phi(x) = x$*):*

$$Q^\star_{ij} = R_{ij} - \nu_i - \mu_j - [S_j^\top D^\top \alpha_j]_i \tag{4}$$

The solution is additive *in dual potentials.*

(b) ***KL*** *(*$\phi(x) = x \log x - x$*, so* $\nabla\phi(x) = \log x$*):*

$$Q^\star_{ij} = R_{ij} \exp\left(-\nu_i - \mu_j - [S_j^\top D^\top \alpha_j]_i\right) \tag{5}$$

The solution is multiplicative *in dual potentials.*

Proof. The Lagrangian for (2) is:

$$\begin{aligned}\mathcal{L}(Q, \nu, \mu, \alpha) = D_\phi(Q, R) + \lambda V_A(Q) \\ + \sum_i \nu_i \left(\sum_j Q_{ij} - 1\right) + \sum_j \mu_j \left(\sum_i Q_{ij} - T_j\right)\end{aligned} \tag{6}$$

Taking the derivative with respect to Q_{ij} and using the subgradient of $[\cdot]_+$:

$$\frac{\partial \mathcal{L}}{\partial Q_{ij}} = \nabla\phi(Q_{ij}) - \nabla\phi(R_{ij}) + \nu_i + \mu_j + [S_j^\top D^\top \alpha_j]_i = 0$$

Rearranging yields (3). The special cases follow by substituting $\nabla\phi(x) = x$ (Euclidean) and $\nabla\phi(x) = \log x$ (KL). □

7.2 KL Isotonic Projection with Sum Constraint

The column subproblem for KL calibration requires projecting onto the intersection of the isotonic cone and a hyperplane (sum constraint). Unlike the Euclidean case which uses arithmetic mean pooling and additive shifts, KL requires geometric mean pooling and multiplicative scaling.

Lemma 7.2 (KL Column Projection)**.** *For KL divergence, the column subproblem*

$$\min_{x_1 \le \cdots \le x_N,\, \sum_i x_i = T} \sum_{i=1}^{N} \left[x_i \log \frac{x_i}{w_i} - x_i + w_i \right] \tag{7}$$

has solution obtained by:

1. *Apply generalized PAV with* ***geometric mean*** *pooling*
2. *Apply* ***multiplicative rescaling****:* $x^\star = y \cdot (T / \sum_i y_i)$

Contrast with Euclidean: PAV (arithmetic mean) + additive shift.

Proof. **Step 1: Geometric Mean Pooling.**

Consider a block B of consecutive indices that must be pooled to satisfy isotonicity. The KL-optimal constant value z_B for this block minimizes:

$$\sum_{i \in B} \left[z_B \log \frac{z_B}{w_i} - z_B + w_i \right]$$

Taking the derivative with respect to z_B:

$$\frac{\partial}{\partial z_B} = \sum_{i \in B} [\log z_B - \log w_i + 1 - 1] = |B| \log z_B - \sum_{i \in B} \log w_i = 0$$

Solving:

$$\log z_B = \frac{1}{|B|} \sum_{i \in B} \log w_i \quad \Rightarrow \quad z_B = \exp\left(\frac{1}{|B|} \sum_{i \in B} \log w_i \right) = \left(\prod_{i \in B} w_i \right)^{1/|B|}$$

This is the **geometric mean**, contrasting with the arithmetic mean for Euclidean.

Step 2: Multiplicative Rescaling.

After PAV, let $y = (y_1, \ldots, y_N)$ be the isotonic solution. To satisfy $\sum_i x_i = T$, we seek a scalar c such that $\sum_i c \cdot y_i = T$. Thus $c = T / \sum_i y_i$ and $x_i^\star = c \cdot y_i$.

This multiplicative scaling preserves isotonicity since the isotone cone $\{x : x_1 \leq \cdots \leq x_N\}$ is closed under positive scalar multiplication.

For Euclidean, the optimal adjustment is additive: $x_i^\star = y_i + (T - \sum_i y_i)/N$. □

7.3 Anchor-Reference Decoupling

A key innovation in our KL framework is the separation of two roles:

- **Reference** R: The distribution from which we measure KL divergence.
- **Anchor** A: The distribution that determines rank orderings to preserve.

Definition 7.3 (Decoupled KL Calibration)**.** *Given input P, reference R, anchor A, and target marginals T:*

$$Q^\star = \arg\min_{Q \in \mathcal{C}(T)} KL(Q \| R) \quad \textit{s.t.} \quad Q_{\cdot j} \textit{ is isotonic w.r.t. } A_{\cdot j} \textit{ ordering} \tag{8}$$

When $R = A = P$, this reduces to standard rank-preserving calibration. Decoupling enables scenarios where:

- We want to measure divergence from one model (R) while preserving rankings from another (A) — e.g., ensemble calibration.
- The reference R is a prior or historical distribution, while A comes from the current classifier.
- Label shift adjustment where R reflects source domain and A reflects target domain ordering preferences.

7.4 Pareto Frontier Characterization

Rather than reporting a single solution for a tuned λ, we advocate computing the full Pareto frontier of KL divergence versus rank violation.

Proposition 7.4 (Pareto Monotonicity)**.** *For the soft problem* (2)*, let* $Q^\star(\lambda)$ *denote the solution for penalty parameter* λ*. Then:*

1. $V_A(Q^\star(\lambda))$ *is non-increasing in* λ*.*
2. $KL(Q^\star(\lambda)\|R)$ *is non-decreasing in* λ*.*

Thus $\{(KL(Q^\star(\lambda)\|R), V_A(Q^\star(\lambda))) : \lambda \geq 0\}$ *traces the Pareto frontier.*

Proof. Standard results from multi-objective optimization: increasing λ places more weight on rank violation, driving V_A down at the cost of increased KL divergence from the reference. □

The Pareto frontier allows practitioners to choose an operating point based on their relative preference for calibration accuracy versus rank preservation, rather than relying on a single "optimal" λ.

8 Computational Considerations

Three implementation choices are worth noting. First, for each class j we compute and cache the permutation that sorts $P_{\cdot j}$ once; all subsequent iterates reuse this order. Second, for the row projection we use the $O(J \log J)$ sorter-based simplex projection [Duchi et al., 2008], though Condat's variant [Condat, 2016] is a drop-in alternative with strong empirical performance. Third, both Dykstra and ADMM benefit from warm starts that reuse PAV block structure and previous scalar shifts.

9 Experiments

We evaluate the method on two complementary benchmarks: a synthetic test where the ground-truth target posterior is known exactly, and real-data experiments with shifted class distributions. We report Brier score, ECE (top-label), accuracy, rank-preservation rate (fraction of within-class pairs whose order is unchanged), and column-total deviation $\max_j |\sum_i Q_{ij} - T_j|$ to verify constraint satisfaction.

9.1 Synthetic Test

We construct a 3-class Gaussian mixture where the base model outputs the exact source posterior. The target domain differs only by class priors, so classical logit-shift yields the exact target posterior. This setup directly

Table 1: Synthetic experiment: tradeoff between RMSE to true target posterior and rank preservation. Logit-shift achieves near-zero RMSE but breaks ranks; strict RPC preserves ranks exactly but deviates from the optimal posterior.

Method	RMSE to Target	Rank Rate	Max Violation
original	0.0805	1.000	0.0000
logit_shift	0.0000	0.977	0.2754
rpc_strict	0.2646	1.000	0.0000
rpc_eps_0.01	0.2459	0.867	0.0100
rpc_eps_0.05	0.2146	0.836	0.0500

tests the fundamental tradeoff from Section 5.1: logit-shift is KL-optimal but may scramble ranks; our method preserves ranks but deviates from the KL-optimal posterior.

Figure 1 shows the tradeoff frontier. Logit-shift achieves near-zero RMSE to the true target posterior but breaks ranks. Strict RPC preserves ranks exactly but deviates from the optimal posterior. The ϵ-slack variants interpolate smoothly between the two.

9.2 Real-Data Shifted Deployment

We train 300-tree random forests [Pedregosa et al., 2011] on UCI `digits` and `wine` datasets, then create shifted holdout sets by resampling with altered class composition. We use exact target counts as T and average over 5 random seeds.

Figure 2 shows the Brier-rank tradeoff across methods. Original probabilities preserve ranks by construction but fail to match target totals. Logit-shift achieves the best Brier score but breaks ranks and falls short of exact total matching (column error > 0). RPC methods achieve exact total matching (column error $= 0$), with RPC-strict additionally preserving ranks exactly where it converges.

10 Discussion and Limitations

The approach is suitable when relative ordering matters, as in regulatory triage, survey weighting, or fairness-constrained ranking. It is less suitable when the base model has poor discrimination (no meaningful ranks to preserve) or when computational budgets prohibit iterative projections on very

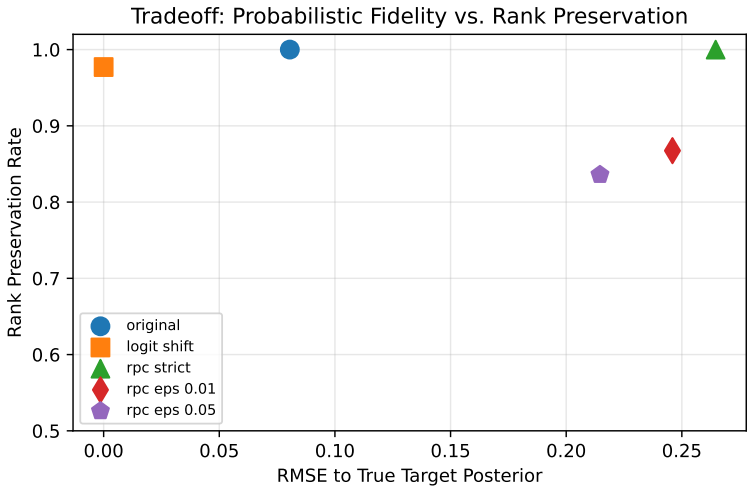

Figure 1: Synthetic tradeoff: RMSE to true target posterior vs. rank preservation rate. Logit-shift achieves near-zero RMSE but scrambles ranks; strict RPC preserves ranks exactly but deviates from the true posterior; ϵ-slack variants interpolate.

large N, where stochastic variants may help.

The choice between methods depends on the downstream use. When predictions will be used to rank instances within classes, as in prioritization queues or risk rankings, preserving ranks is essential, and strict RPC is appropriate. When small rank violations are acceptable, the ϵ-slack variants offer a smooth tradeoff, with ϵ tunable against a validation objective (Brier, ECE) or a domain-specific maximum tolerable violation. When the application cares only about expected loss minimization and not about individual instance comparisons, classical multiplicative reweighting (logit-shift) remains the right tool, with the understanding that ranks may not be preserved.

11 Conclusion

We presented a projection-based multiclass calibrator that preserves within-class rankings while matching population totals. The method reduces to PAV plus a scalar shift inside Dykstra or ADMM, is efficient in practice, and integrates into existing calibration pipelines.

Table 2: Real-data experiments with shifted class distributions. Results averaged over 5 random seeds. RPC-strict achieves exact column totals and rank preservation where it converges; on `digits` (10 classes with extreme shift), strict convergence fails and practitioners should use the nearly-isotonic variant.

Dataset	Method	Brier ↓	ECE ↓	Rank Rate ↑	Col Err ↓
digits	logit_shift	0.156±0.036	0.134±0.038	0.922±0.020	49.18±16.54
digits	original	0.125±0.021	0.221±0.027	1.000±0.000	89.62±24.99
digits	rpc_eps_0.02	0.254±0.099	0.236±0.050	0.770±0.022	0.00±0.00
digits	rpc_strict	—	—	—	—
wine	logit_shift	0.051±0.046	0.064±0.048	0.977±0.013	3.14±2.30
wine	original	0.049±0.013	0.121±0.025	1.000±0.000	6.64±2.62
wine	rpc_eps_0.02	0.097±0.083	0.082±0.057	0.860±0.057	0.00±0.00
wine	rpc_strict	0.101±0.087	0.079±0.052	1.000±0.000	0.00±0.00

References

Alon Arad and Saharon Rosset. Improving multi-class calibration through normalization-aware isotonic techniques. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. URL `https://arxiv.org/abs/2512.09054`.

Heinz H Bauschke and Jonathan M Borwein. Dykstra's alternating projection algorithm for two sets. *Journal of Approximation Theory*, 79(3): 418–443, 1994.

Stephen Boyd, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. Distributed optimization and statistical learning via the alternating direction method of multipliers. *Foundations and Trends in Machine Learning*, 3(1):1–122, 2011.

Laurent Condat. Fast projection onto the simplex and the ℓ_1 ball. *Mathematical Programming*, 158(1-2):575–585, 2016. doi: 10.1007/s10107-015-0946-6.

John Duchi, Shai Shalev-Shwartz, Yoram Singer, and Tushar Chandra. Efficient projections onto the l1-ball for learning in high dimensions. In *Proceedings of the 25th International Conference on Machine Learning*, pages 272–279, 2008.

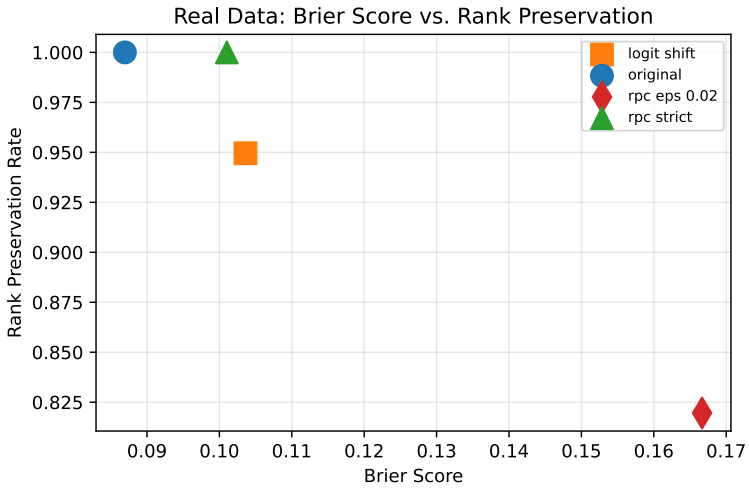

Figure 2: Real data: Brier score vs. rank preservation rate across methods, averaged across both datasets and all seeds.

Richard L Dykstra. An algorithm for restricted least squares regression. *Journal of the American Statistical Association*, 78(384):837–842, 1983.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, pages 1321–1330, 2017.

Meelis Kull, Miquel Perello-Nieto, Markus Kängsepp, Telmo Silva Filho, Hao Song, and Peter Flach. Beyond temperature scaling: Obtaining well-calibrated multiclass probabilities with Dirichlet calibration. In *Advances in Neural Information Processing Systems*, pages 12316–12326, 2019.

Zachary C. Lipton, Yu-Xiang Wang, and Alexander J. Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018. URL `https://arxiv.org/abs/1802.03916`.

Xingchen Ma and Matthew B. Blaschko. Meta-cal: Well-controlled post-hoc calibration by ranking. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. URL `https://arxiv.org/abs/2105.04290`.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer,

Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002. doi: 10.1162/089976602753284446.

Ryan J Tibshirani. Dykstra's algorithm, ADMM, and coordinate descent: connections, insights, and extensions. *Advances in Neural Information Processing Systems*, 30, 2017.